NFC Book V Branch Legitimacy L4 Shell

Schema v0.39.0

L4 Spine Release Candidate

Universal Branch Legitimacy Theorem (UBLT) — closes branch-spine dependency gap

Generated by NFC Governance Engine v39

Contents

Book V closes the branch-spine dependency gap. The UBLT (Thm. V.8.1) is cited by all 10 branch books. With Book V machine records, the governance engine can trace the full branch-to-spine dependency chain. This is an L4 spine-pilot shell — the canonical Book V is the authoritative source.

Book V Branch Legitimacy — L4 Generated Shell

Book V is the most branch-facing spine book. Its Universal Branch Legitimacy Theorem (UBLT, Thm. V.8.1) is cited by every branch book in the corpus. The L4 pilot closes the machine-readable dependency gap between the spine and the branch books.

Primary Unconditional Results

Machine ID	St.	Canonical label
`prop:V.branch-projection`	[U]	Branch Projection Theorem (Prop.~V.3.1) [U]
`thm:V.UBLT`	[U]	Universal Branch Legitimacy Theorem (UBLT / Thm.~V.8.1) [U]
`cor:V.mandatory-declarations`	[U]	Mandatory Branch Declarations [U]
`cor:V.open-branch-still-canonical`	[U]	Open Branch Still Canonical [U]
`thm:V.collapse`	[U]	Collapse Criterion [U]
`thm:V.diamond`	[U]	Diamond Confluence Principle [U]

The Five UBLT Conditions

1. Licensed intake: $(\mathcal{C}, F)$ is a licensed branch candidate
2. Certified source descent: $E = F(\mathbf{U})$ from the universal source
3. Endpoint visibility: E is branch-visible (invariant under source equivalence)
4. Legitimacy witness: a legitimacy witness for $(\mathcal{C}, F)$ exists
5. No hidden-channel supplementation: no undeclared structure enters E

The UBLT says: lawful branch $\Leftrightarrow$ all five conditions hold.

Full Claim Catalog

Machine ID	Kind	St.	Title
`sr:V.source-descent-only`	standing_r	[D]	Source-Descent-Only Rule
`sr:V.no-hidden-supplementation`	standing_r	[D]	No-Hidden-Supplementation Rule
`sr:V.endpoint-visibility`	standing_r	[D]	Endpoint-Visibility Standing Rule
`sr:V.legitimacy-before-closure`	standing_r	[D]	Legitimacy-Before-Closure Standing Rule
`def:V.branch-candidate`	definition	[D]	Branch Candidate
`def:V.endpoint-datum`	definition	[D]	Endpoint Datum
`def:V.hidden-channel`	definition	[D]	Hidden Channel
`def:V.legitimacy-witness`	definition	[D]	Legitimacy Witness
`def:V.diamond-confluence`	definition	[D]	Diamond Confluence Configuration
`prop:V.branch-projection`	propositio	[U]	Branch Projection Theorem (Prop. V.3.1)
`prop:V.endpoint-visibility`	propositio	[U]	Endpoint Visibility Proposition
`prop:V.hidden-channel-exclusion`	propositio	[U]	Hidden Channel Exclusion Proposition
`thm:V.collapse`	theorem	[U]	Collapse Criterion
`thm:V.diamond`	theorem	[U]	Diamond Confluence Principle
`thm:V.UBLT`	theorem	[U]	Universal Branch Legitimacy Theorem (UBLT, Thm. V.
`cor:V.mandatory-declarations`	corollary	[U]	Mandatory Branch Declarations Corollary
`cor:V.open-branch-still-canonical`	corollary	[U]	Open Branch Still Canonical Corollary
`thm:V.screened-sector`	theorem	[C]	Screened Sector as Canonical Branch-Visible Endpoi
`status:V`	remark	[R]	Book V Status — UBLT unconditional; screened secto

Branch-Spine Dependency Linkage

The following branches cite `thm:V.UBLT` (Thm. V.8.1): `GR`, `YM`, `NS`, `SM`, `BIO`, `LING`, `CRYST`, `SCC`, `SPEC`, `RH`. With Book V machine records in place, the governance engine can now trace the full branch-to-spine dependency chain, for example:

`thm:GR.domain` → `thm:VII.6.4` → `thm:V.UBLT` → `def:V.branch-candidate` → ···